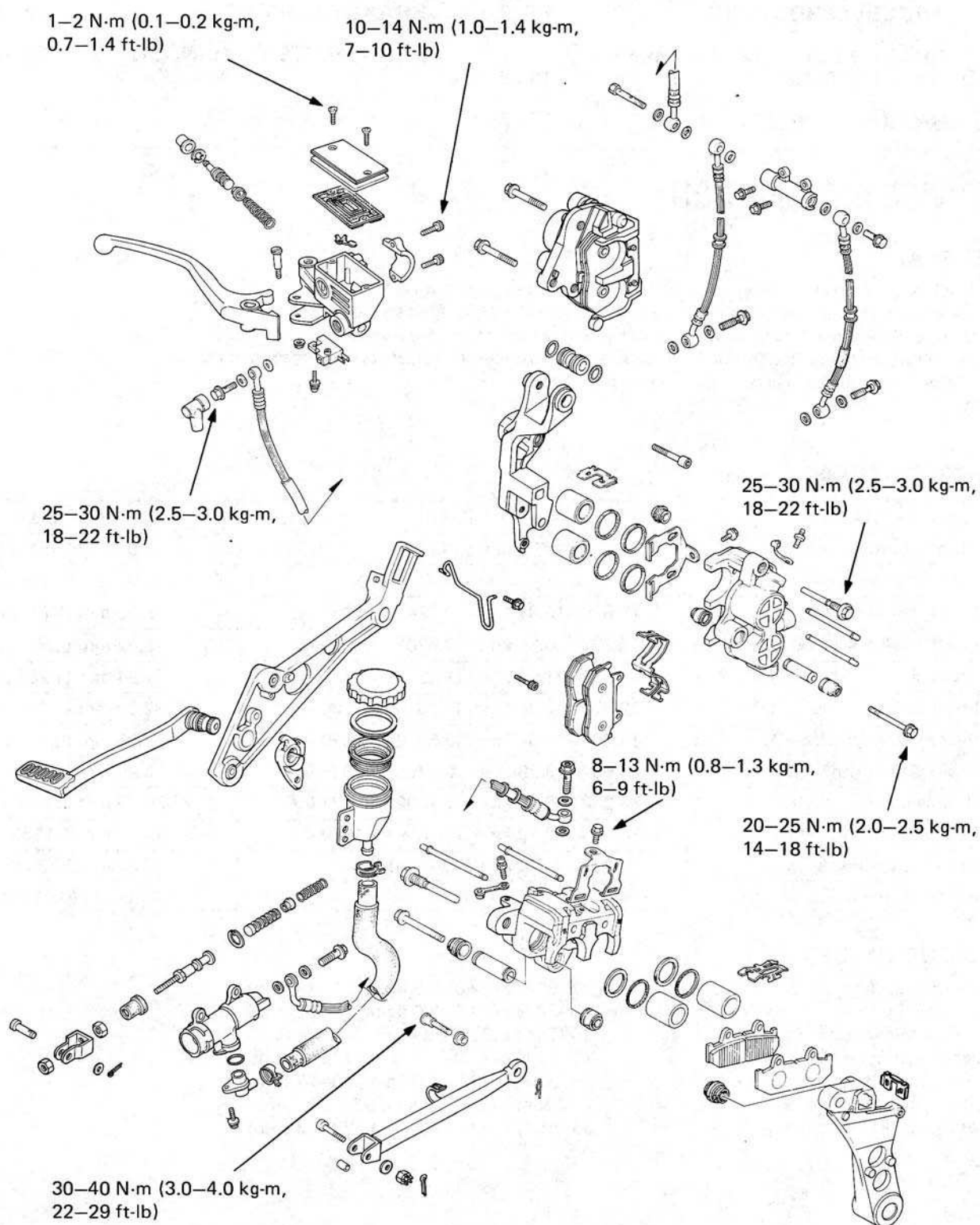




HONDA CBX750F

15. Hydraulic Brakes





SERVICE INFORMATION	15-1	FRONT MASTER CYLINDER	15-8
TROUBLESHOOTING	15-2	BRAKE CALIPERS	14-11
BRAKE FLUID REPLACEMENT/ AIR BLEEDING	15-3	REAR MASTER CYLINDER	15-14
BRAKE PAD/DISC	15-5		

SERVICE INFORMATION

GENERAL

- The brake calipers can be removed without disconnecting the hydraulic system.
- Bleed the hydraulic system if it is disassembled or if the brake feels spongy.
- Do not allow foreign material to enter the system when filling the reservoir.
- Avoid spilling brake fluid on painted surfaces or instrument lenses, as severe damage can result.
- Always check brake operation before riding the motorcycle.

SPECIFICATIONS

ITEM	STANDARD	SERVICE LIMIT
Front disc thickness	5.0 mm (0.197 in)	4.0 mm (0.157 in)
Front disc runout	-	0.30 mm (0.012 in)
Front master cylinder I.D.	15.870-15.913 mm (0.6248-0.6265 in)	15.93 mm (0.627 in)
Front master piston O.D.	15.827-15.854 mm (0.6231-0.6242 in)	15.82 mm (0.623 in)
Front caliper piston O.D.	31.948-31.998 mm (1.2578-1.2598 in)	31.94 mm (1.257 in)
Front caliper cylinder I.D.	32.030-32.080 mm (1.2610-1.2630 in)	32.09 mm (1.263 in)
Rear master cylinder I.D.	14.000-14.043 mm (0.5512-0.5529 in)	14.06 mm (0.554 in)
Rear master piston O.D.	13.957-13.984 mm (0.5495-0.5506 in)	13.95 mm (0.549 in)
Rear caliper cylinder I.D.	30.230-30.280 mm (1.1902-1.1921 in)	30.29 mm (1.193 in)
Rear caliper piston O.D.	30.148-30.198 mm (1.1869-1.1889 in)	30.14 mm (1.1866 in)
Rear disc thickness	7.0 mm (0.276 in)	6.0 mm (0.236 in)
Rear disc runout	-	0.30 mm (0.012 in)

TORQUE VALUES

Caliper mount bolt	20-25 N.m (2.0-2.5 kg.m, 14-18 ft.lb)
Caliper pin bolt	25-30 N.m (2.5-3.0 kg.m, 18-22 ft.lb)
Pad pin retainer bolt	8-13 N.m (0.8-1.3 kg.m, 6-9 ft.lb)
Front master cylinder holder	10-14 N.m (1.0-1.4 kg.m, 7-10 ft.lb)
Brake hose oil bolt	25-30 N.m (2.5-3.0 kg.m, 18-22 ft.lb)
Master cylinder reservoir cap	1-2 N.m (0.1 -0.2 kg.m, 0.7-1.4 ft.lb)
Footpeg bracket mount bolt	35-45 N.m (3.5-4.5 kg.m, 25-33 ft.lb)

TOOL

Special

Snap ring pliers	07914-3230001
------------------	---------------



TROUBLESHOOTING

Brake lever/pedal soft or spongy

1. Air bubbles in hydraulic system
2. Low fluid level
3. Hydraulic system leaking

Brake lever/pedal too hard

1. Sticking piston(s)
2. Clogged hydraulic system
3. Pads glazed or worn excessively

Brake drag

1. Hydraulic system sticking
2. Sticking piston(s)

Brakes grab

1. Pads contaminated
2. Disc or wheel misaligned

Brake chatter or squeal

1. Pads contaminated
2. Excessive disc runout
3. Caliper installed incorrectly
4. Disc or wheel misaligned



BRAKE FLUID REPLACEMENT/ AIR BLEEDING

Check the fluid level with the fluid reservoir parallel to the ground.

CAUTION

Avoid spilling fluid on painted surfaces. Place clean shop towels over the fuel tank whenever the system is being serviced.

BRAKE FLUID DRAINING

Remove the reservoir cap and diaphragm. Connect a bleed hose to the bleed valve to avoid spilling fluid.

WARNING

A brake fluid contaminated brake disc or pad reduces stopping power. Discard contaminated pads and clean a contaminated disc with a high quality brake degreasing agent.

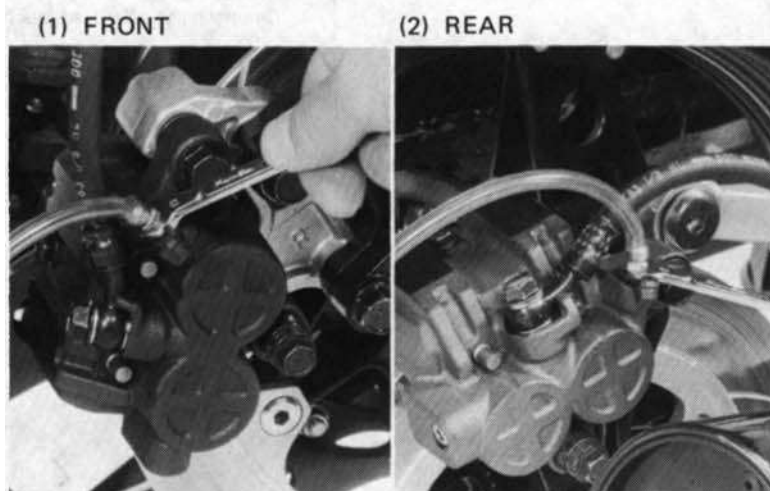
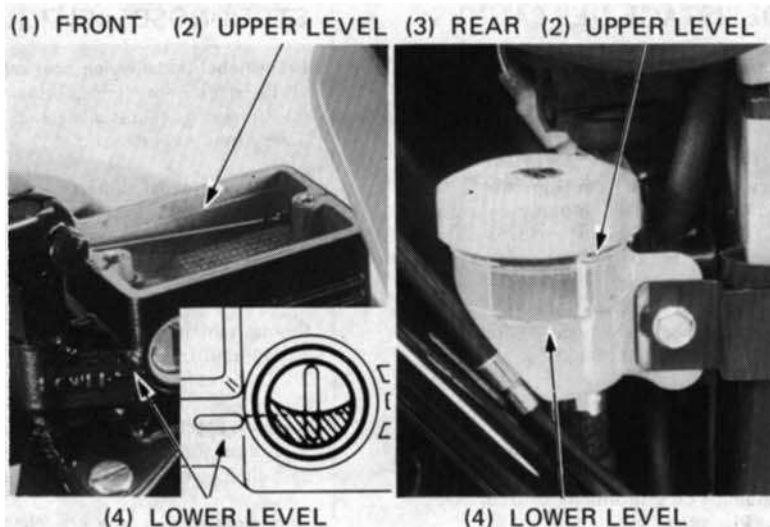
Loosen the caliper bleed valve and pump the brake lever.
Stop operating the lever when fluid stops flowing out of the bleed valve.

BRAKE FLUID FILLING

NOTE:

Do not mix different types of fluid since they are not be compatible.

Close the bleed valve and fill the reservoir. Pump up the system pressure with the lever or pedal until there are no air bubbles in the fluid flowing out of the reservoir small hole and lever or pedal resistance is felt.





15. Hydraulic Brakes

NOTE:

- Check the fluid level often while bleeding the brakes to prevent air from being pumped into the system.
- Use only DOT 4 brake fluid from a sealed container.
- Do not mix brake fluid types and never reused the contaminated fluid which has been pumped out during brake bleeding, because this will impair the efficiency of the brake system.

1. Squeeze the brake lever (or depress the brake pedal), open the bleed valve 1/2 turn and then close the valve.

NOTE:

Do not release the brake lever (or pedal) until the bleed valve has been closed.

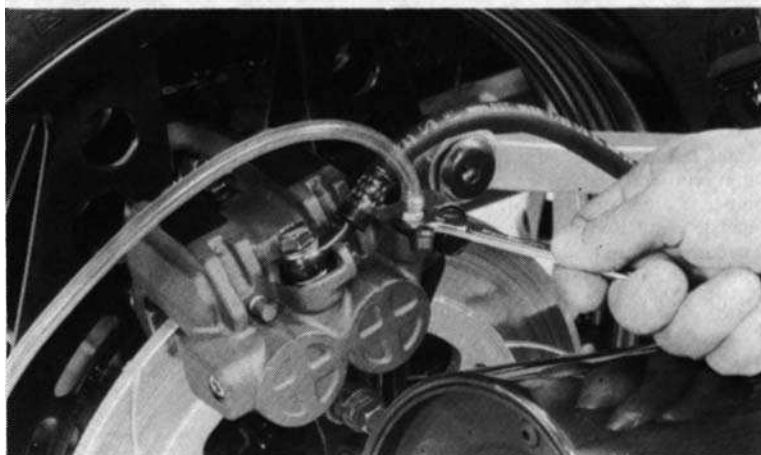
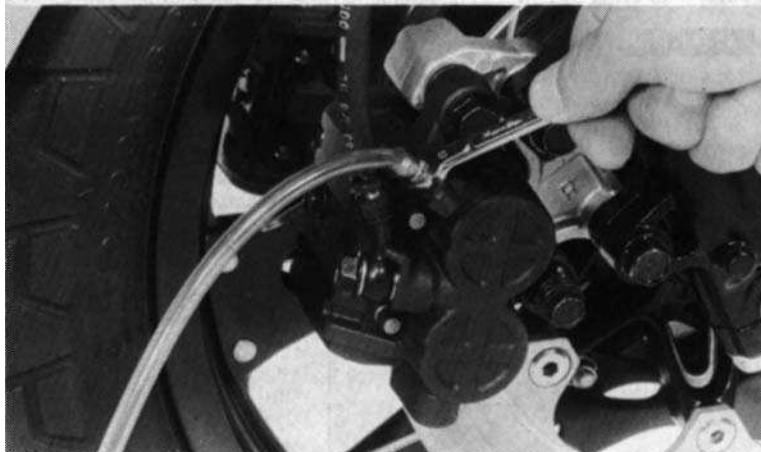
2. Release the brake lever (or pedal) slowly and wait several seconds after it reaches the end of its travel.

Repeat steps 1 and 2 until bubbles cease to appear in the fluid coming out of the bleeder valve.

Fill the fluid reservoir to the upper level mark.

WARNING

A contaminated brake disc or pad reduces stopping power. Discard contaminated pads and clean a contaminated disc with a high quality brake degreasing agent.





BRAKE PAD/DISC

FRONT BRAKE PAD REPLACEMENT

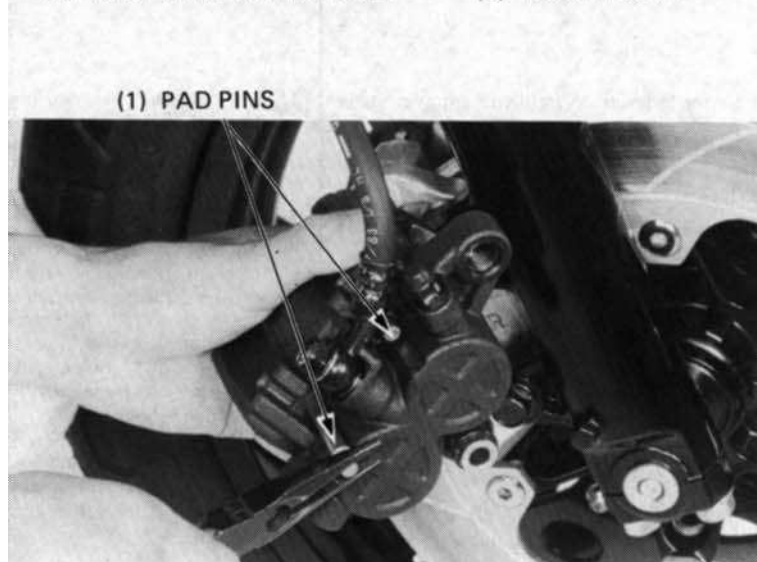
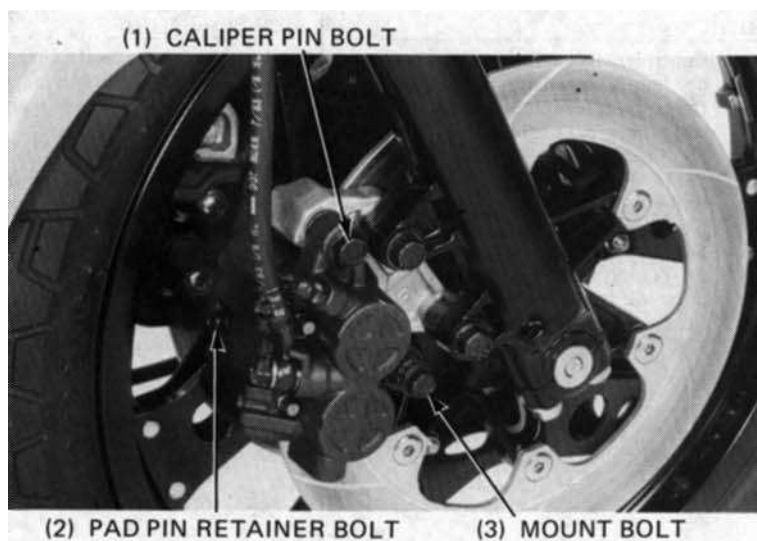
NOTE:

Always replace the brake pads in pairs to assure even disc pressure.

Loosen the pad pin retainer bolt.
Remove the caliper pin bolt and mount bolt.
Remove the caliper from the bracket.

Remove the pad pin retainer and pull the pad pins out of the caliper.
Remove the brake pads.

Position the pad spring in the caliper as shown.
Push the caliper pistons in all the way.

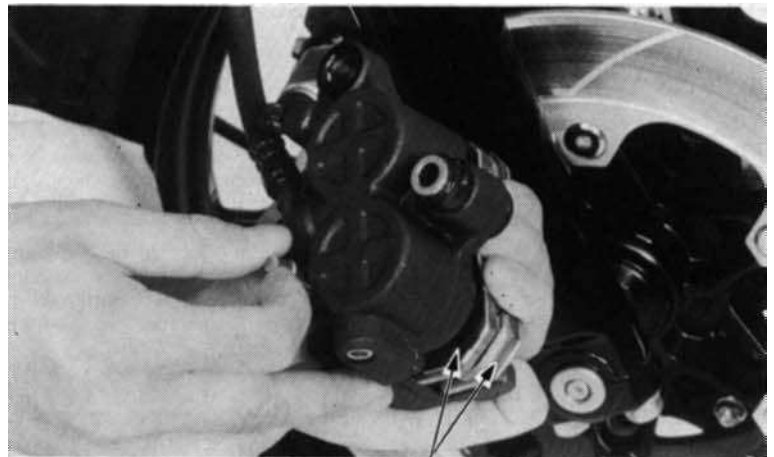




15. Hydraulic Brakes

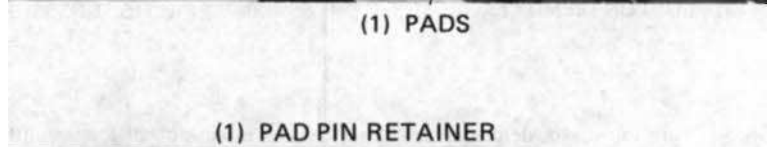
Install the new pads in the caliper.

Install the pad pins, one pad pin first, then install the other pin by pushing the pads against the caliper to depress the pad spring.



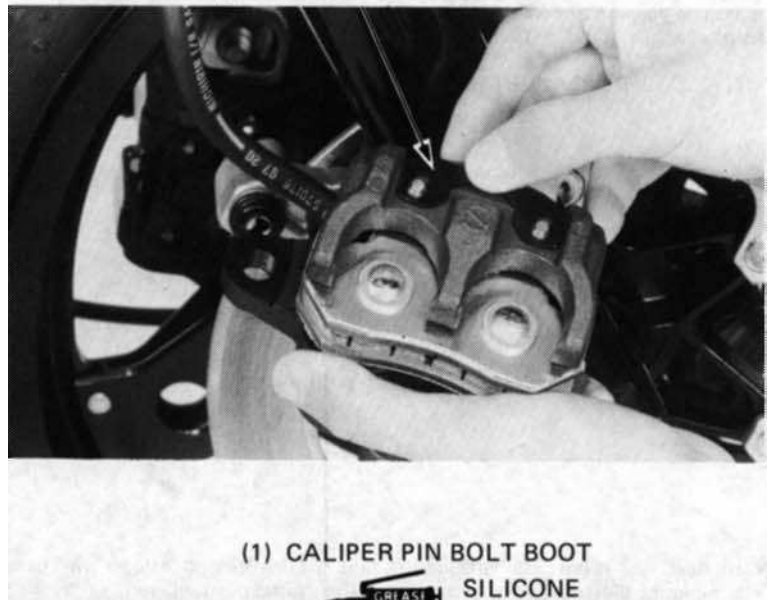
(1) PADS

Place the pad pin retainer over the pad pins. Push the retainer down to secure the pins.



(1) PAD PIN RETAINER

Apply silicone grease to the caliper pin bolt boot and caliper pin bolt. Make sure that the retainer clip is in position on the caliper bracket.



(1) CALIPER PIN BOLT BOOT



(2) RETAINER CLIP



15. Hydraulic Brakes

Install the caliper to the bracket so the disc is positioned between the pads, being careful not to damage the pads.

Tighten the caliper mount bolt and pin bolt.

TORQUE:

CALIPER MOUNT BOLT:

20-25 Nm (2.0-2.5 kg.m, 14-18 ft.lb)

CALIPER PIN BOLT

25-30 Nm (2.5-3.0 kg.m, 18-22 ft.lb)

Tighten the pad pin retainer bolt.

TORQUE: 8-13 Nm (0.8-1.3 kg.m, 6-9 ft.lb)

REAR BRAKE PAD REPLACEMENT

Loosen the pad pin retainer bolt.

Loosen the caliper mount bolt and remove it from the caliper bracket.

Pivot the caliper up out of the way and remove the caliper from the bracket.

Replace the rear brake pads using the same method as used for front brake pad replacement.

Before installing the caliper to the bracket, apply silicone grease to the caliper pin bolt boot and caliper pin bolt, and make sure that the retainer clip is in position on the bracket.

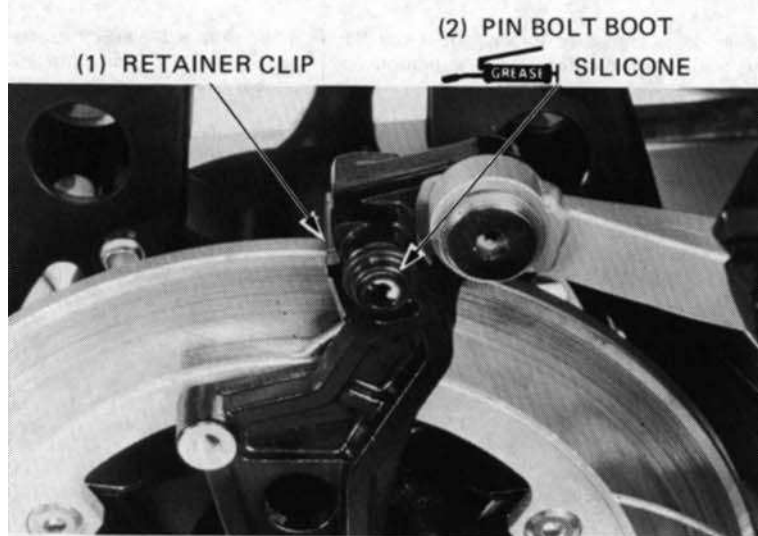
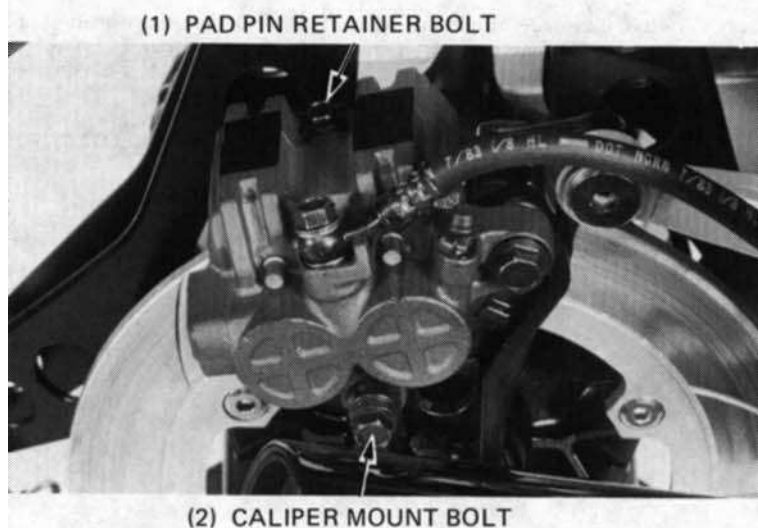
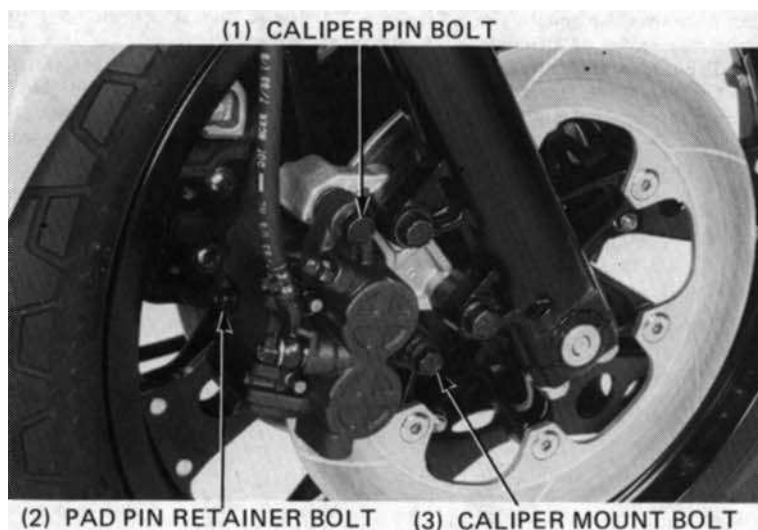
TORQUE:

CALIPER MOUNT BOLT:

20-25 Nm (2.0-2.5 kg.m, 14-18 ft.lb)

PAD PIN RETAINER BOLT:

8-13 Nm (0.8-1.3 kg.m, 6-9 ft.lb)





DISC THICKNESS

Measure the thickness of each disc.

SERVICE LIMIT:

FRONT: 4.0 mm (0.16 in)

REAR: 6.0 mm (0.24 in)

BRAKE DISC WARPAGE

Measure brake disc for warpage.

SERVICE LIMIT: 0.30 mm (0.012 in)

FRONT MASTER CYLINDER

DISASSEMBLY

Drain brake fluid from the hydraulic system.
Remove the brake lever and rear view mirror from the master cylinder.
Disconnect the brake hose.

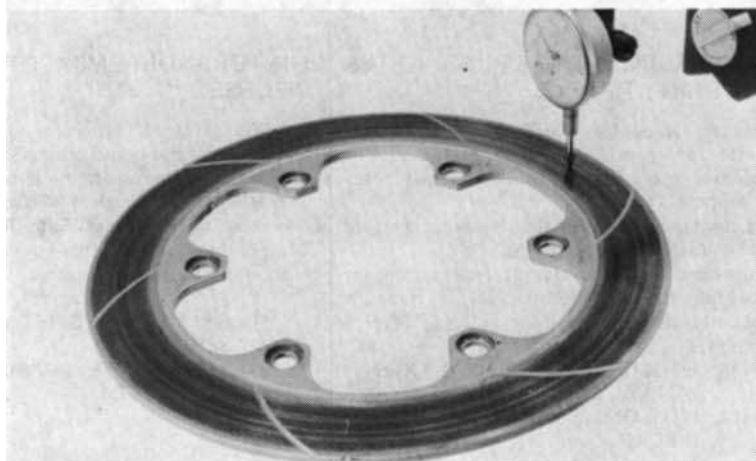
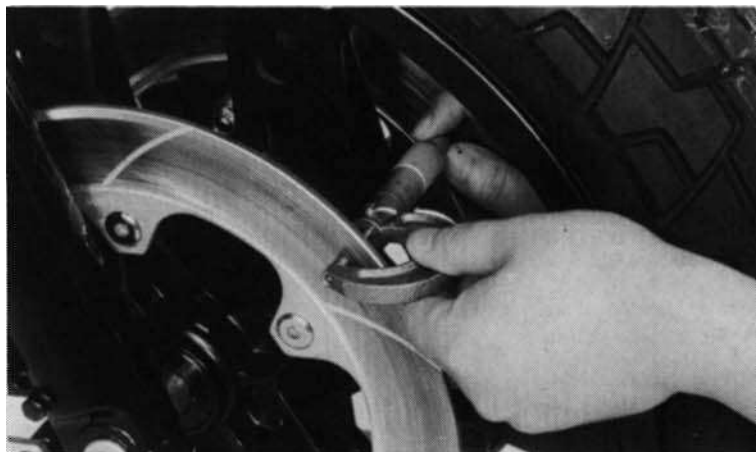
CAUTION

Avoid spilling brake fluid on painted surfaces. Place a rag over the fuel tank whenever the brake system is being serviced.

NOTE:

When removing the fluid hose bolt, cover the end of the hose to prevent contamination. Secure the hose to prevent fluid from leaking out.

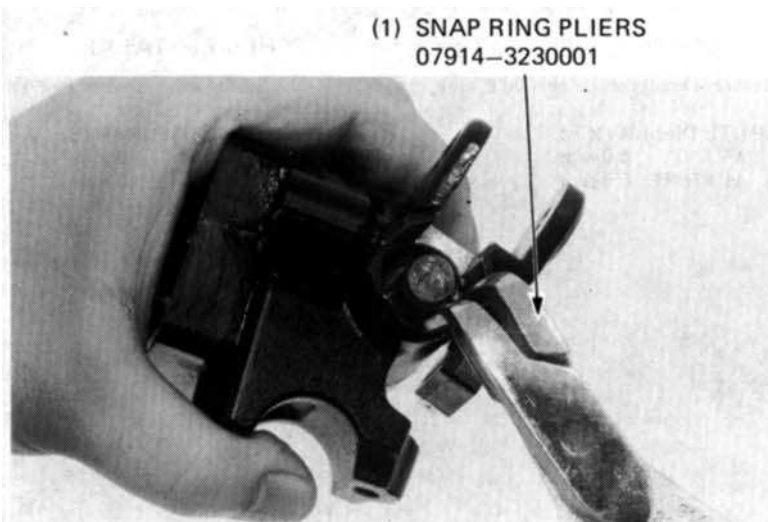
Disconnect the front brake switch wires.
Remove the front brake master cylinder.



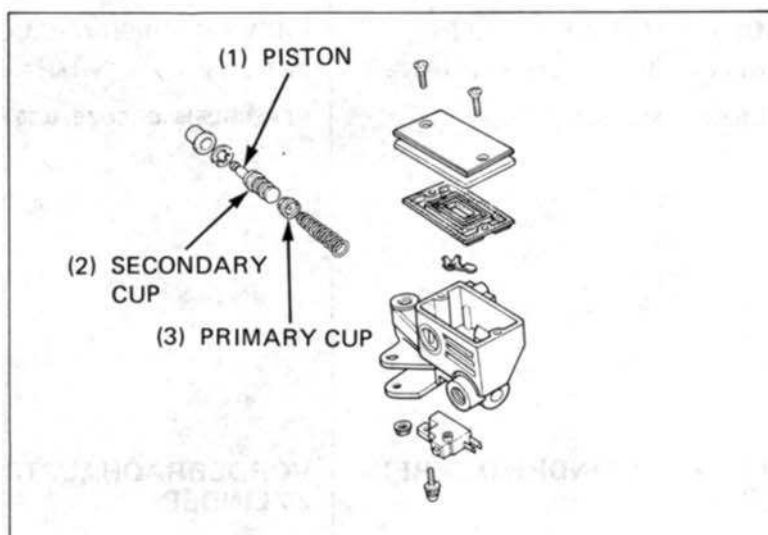


15. Hydraulic Brakes

Remove the piston boot and the snap ring from the master cylinder body.



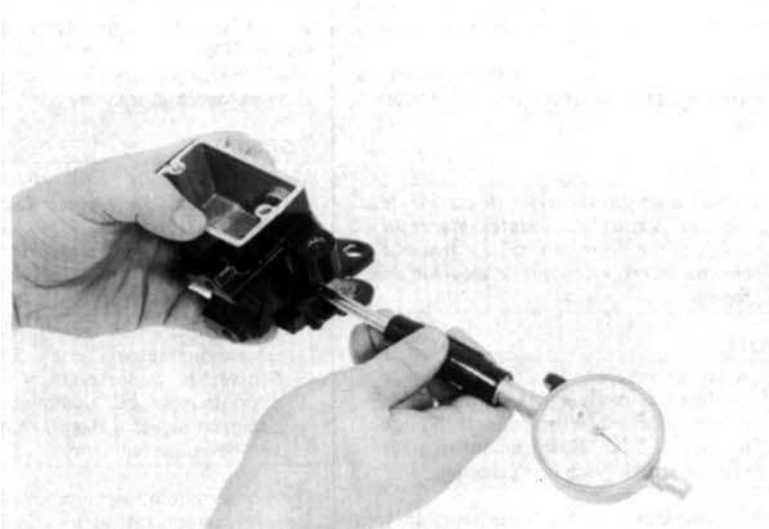
Remove the secondary cup and piston. Then remove the primary cup and spring. Remove the brake light switch from the master cylinder body, if necessary. Clean the inside of the master cylinder and reservoir with brake fluid.



INSPECTION

Measure the master cylinder I.D.
Check the master cylinder for scores, scratches or nicks.

SERVICE LIMIT: 15.93 mm (0.627 in)





HONDA CBX750F

15. Hydraulic Brakes

Measure the master piston O.D.

SERVICE LIMIT: 15.82 mm (0.623 in)

Check the primary and secondary cups for damage before assembly.

ASSEMBLY

CAUTION

Keep the master cylinder piston, cylinder and spring as a set; don't substitute individual parts.

Assemble the master cylinder. Coat the master piston, primary and secondary cups with clean brake fluid before assembly. Install the spring and primary cup together.

Install the piston and snap ring.

CAUTION

Do not allow the lips of the cups to turn inside out and be certain the snap ring is firmly seated in the groove.

Install the boot.

INSTALLATION

Place the front brake master cylinder on the handlebar and install its holder with the "UP" mark facing up. Align the index mark on the holder with the punch mark on the handlebar. Tighten the upper bolt first, then tighten the lower bolt.

TORQUE: 10-14 N.m
(1.0-1.4 kg.m, 7-10 ft.lb)

Install the brake hose with the oil bolt and two sealing washers, and tighten the oil bolt.

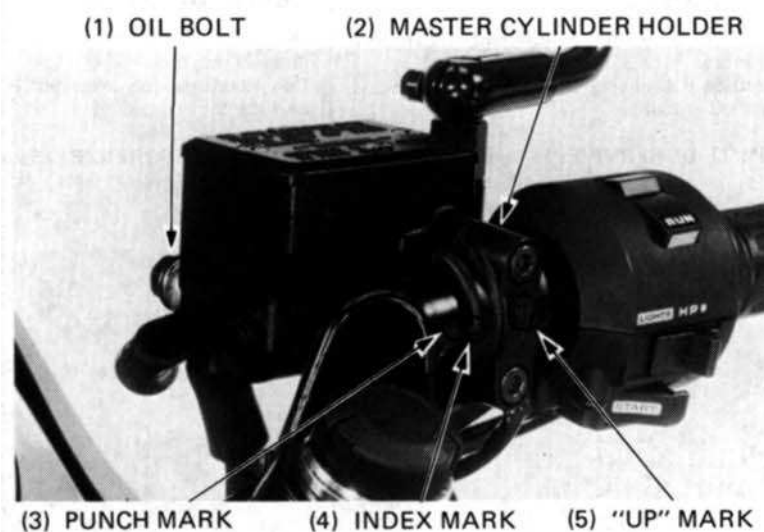
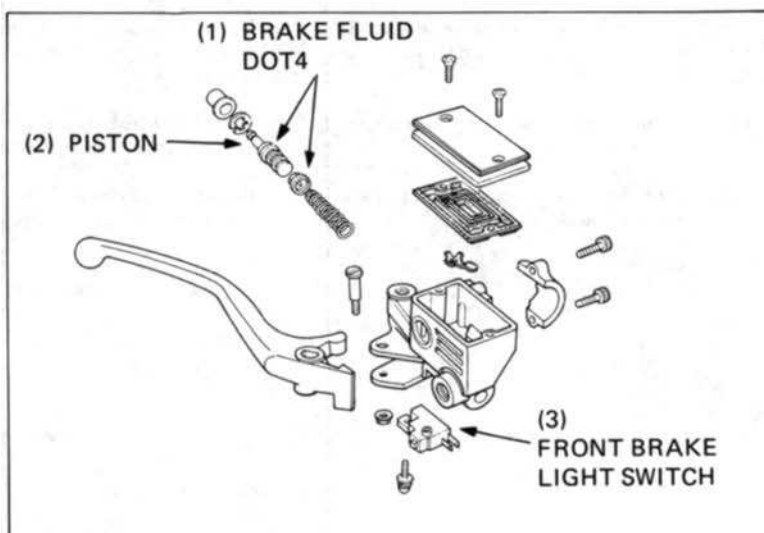
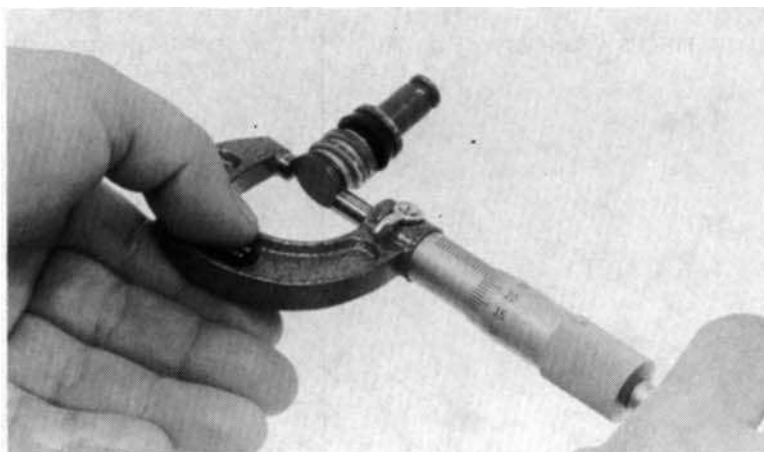
TORQUE: 25-30 N.m
(2.5-3.0 kg.m, 18-22 ft.lb)

Install the brake lever and rear view mirror.

Connect the front brake switch wires.

Fill the reservoir to the upper level and

bleed the front brake system according to page 15-4.





BRAKE CALIPERS

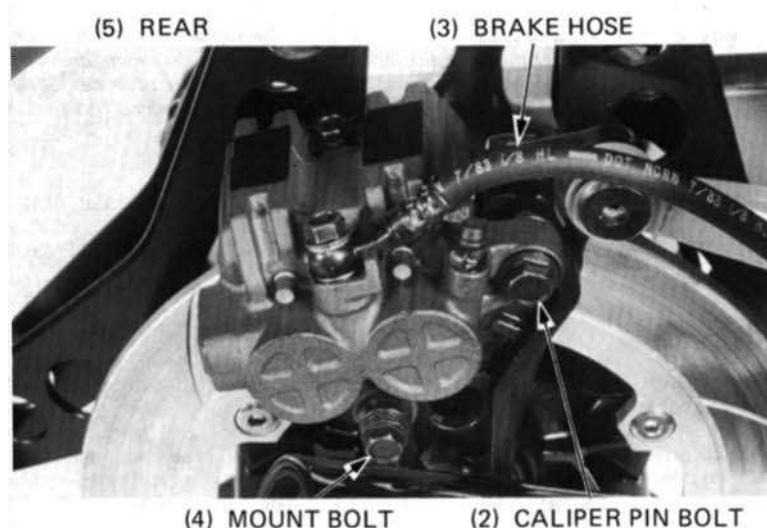
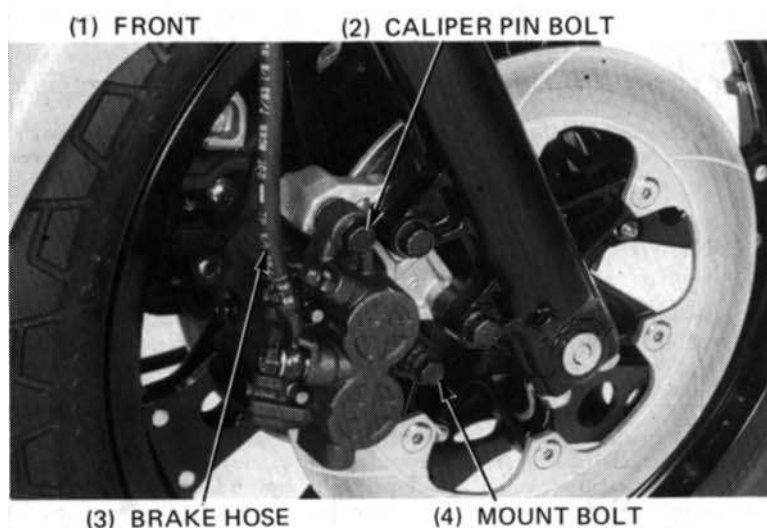
REMOVAL

Place a clean container under the caliper and disconnect the brake hose from the caliper.

CAUTION

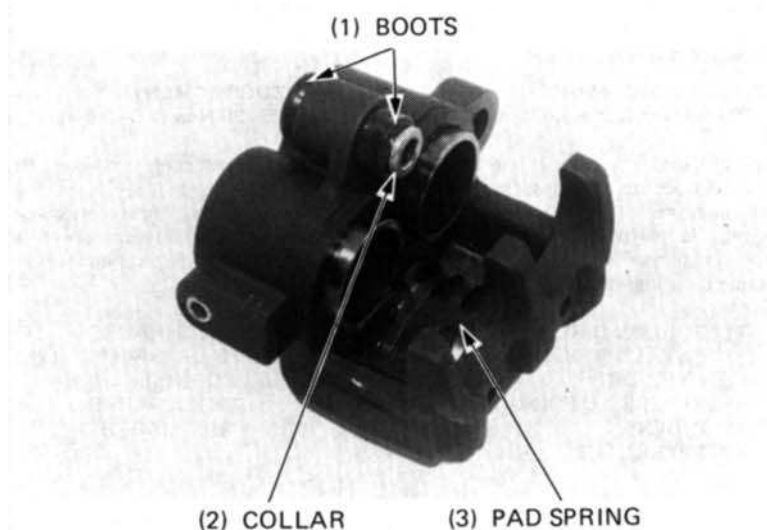
Avoid spilling brake fluid on painted surfaces.

Remove the caliper pin bolt and mount bolt, and remove the caliper.



DISASSEMBLY

Remove the brake pads (page 15-5).
Remove the pad spring.
Remove the caliper pivot collar and boots.
Remove the pistons from the caliper.





HONDA CBX750F

15. Hydraulic Brakes

If necessary, apply compressed air to the caliper fluid inlet to get the piston out. Place a shop rag under the caliper to cushion the piston when it is forced out. Use the air in short spurts.

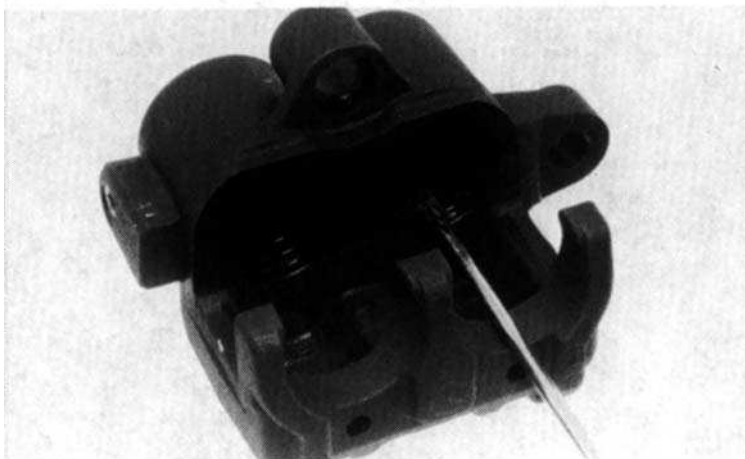
WARNING

Do not bring the nozzle too close to the inlet.

Examine the pistons and cylinders for scoring, scratches or other damage and replace if necessary.



Push the dust and piston seals in and lift them out, then discard them. Clean the piston seal grooves with brake fluid.

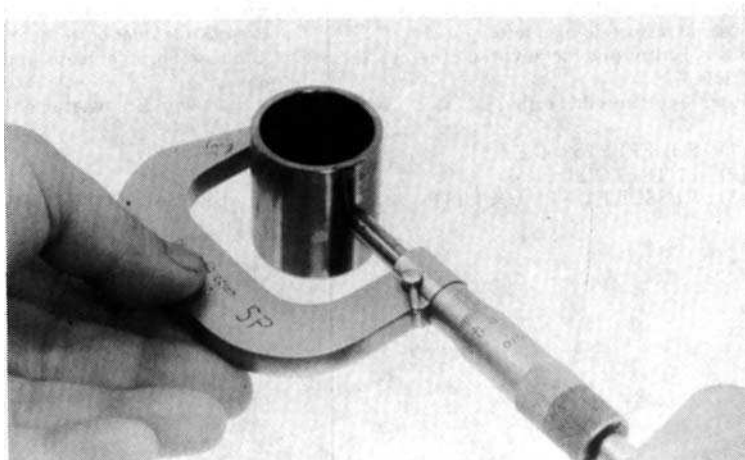


CAUTION

Be careful not to damage the piston sliding surfaces.

PISTON INSPECTION

Check the pistons for scoring, scratches or other damage. Measure the piston diameter with a micrometer.



SERVICE LIMIT:

FRONT: 31.94 mm (1.257 in)

REAR: 30.14 mm (1.1866 in)



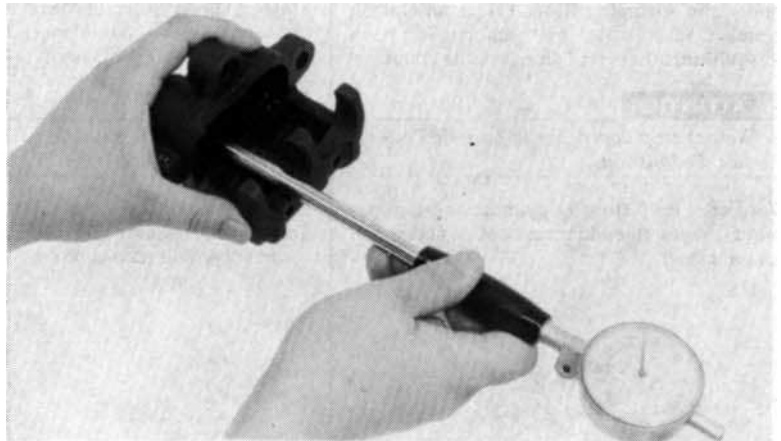
CYLINDER INSPECTION

Check the caliper cylinder for scoring, scratches or other damage. Measure the caliper cylinder bore.

SERVICE LIMIT:

FRONT: 32.09 mm (1.263 in)

REAR: 30.29 mm (1.193 in)



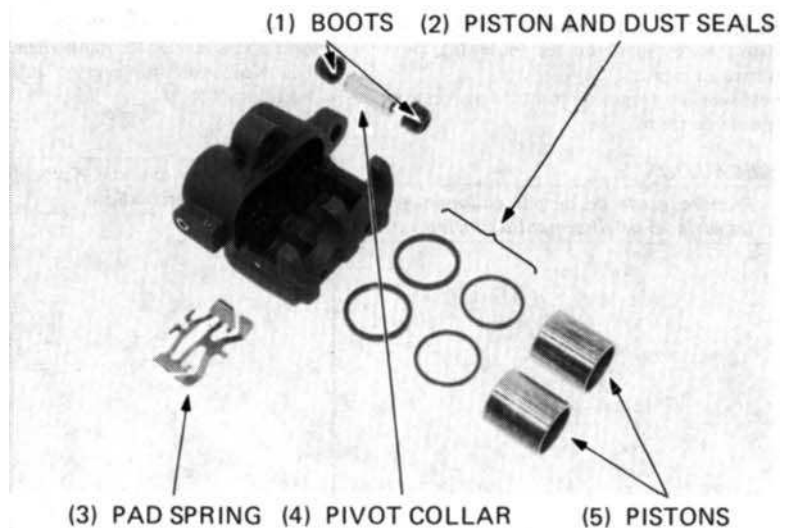
ASSEMBLY

If the pivot collar boots are hardened or deteriorated, replace them with new ones. The piston and dust seals must be replaced with new ones whenever they are removed. Coat the seals with silicone grease or brake fluid before assembly. Install the pistons with the dished ends toward the pads.

Apply silicone grease to the pivot collar and boots and install the collar and boots making sure that the boots are seated in the collar and caliper grooves properly.

Install the pad spring.

Install the pads (page 15-5).

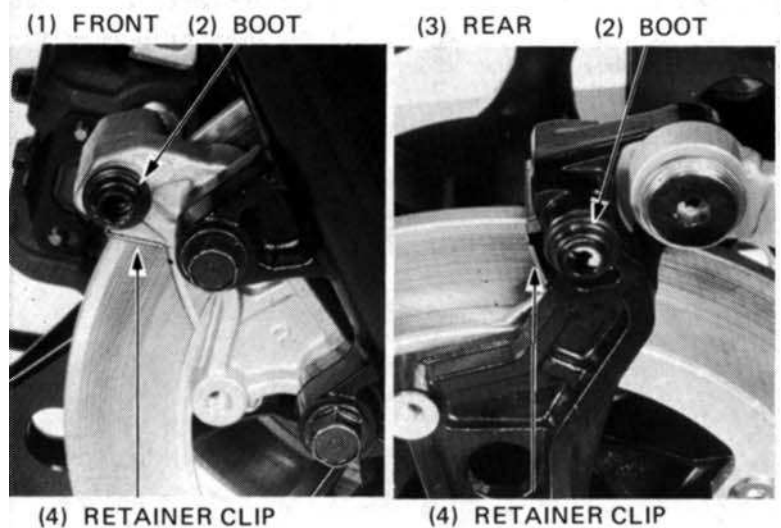


INSTALLATION

Make sure that the retainer clip is in position on the caliper bracket.

Inspect the condition of the caliper pin bolt boot.

Apply silicone grease to the caliper pin bolt boot and bolt.





15. Hydraulic Brakes

Install the caliper assembly to the bracket so that the disc is positioned between the pads, being careful not to damage the pads.

Tighten the caliper mount bolt and pin bolt.

TORQUE:

CALIPER MOUNT BOLT:

20-25 N.m (2.0-2.5 kg.m, 14-18 ft.lb)

CALIPER PIN BOLT:

25-30 N.m (2.5-3.5 kg.m, 18-22 ft.lb)

Connect the brake hose to the caliper with the oil bolt and two sealing washers, and tighten the oil bolt.

TORQUE: 25-30 N.m

(2.5-3.0 kg.m, 18-22 ft.lb)

Fill the brake fluid reservoir and bleed the brake system (page 15-4).

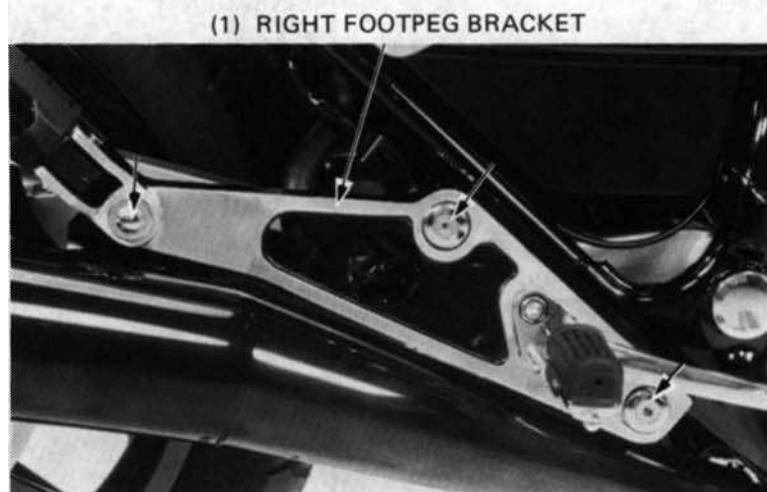
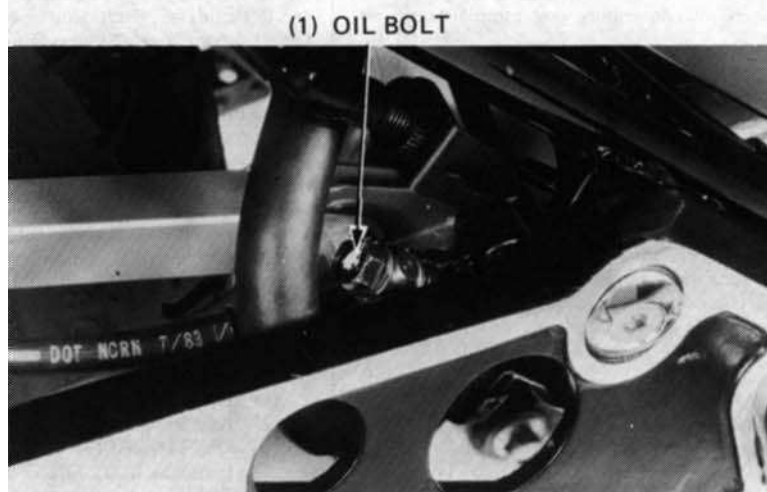
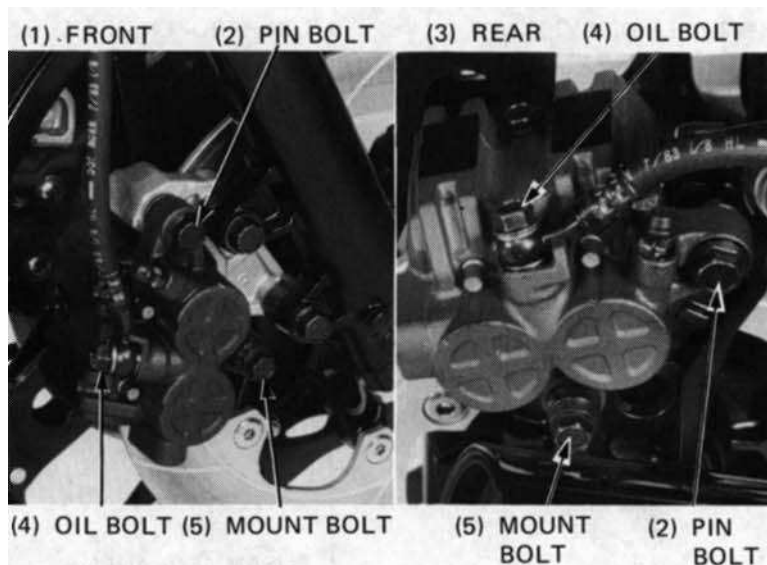
REAR MASTER CYLINDER

REMOVAL

Drain the brake fluid from the rear brake system (page 15-3).

Remove the oil bolt and disconnect the brake hose from the master cylinder.

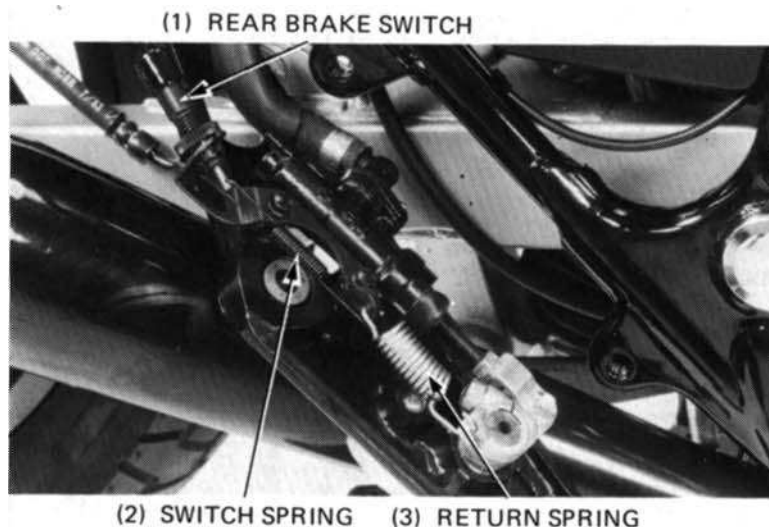
Remove the right footpeg bracket.



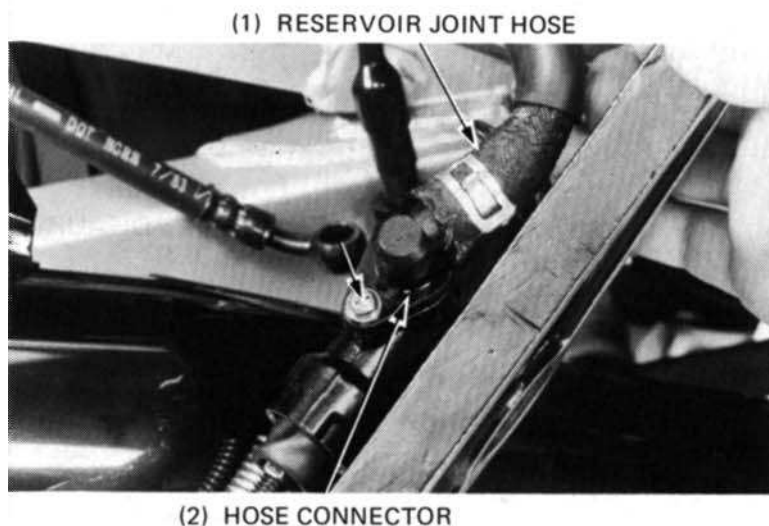


15. Hydraulic Brakes

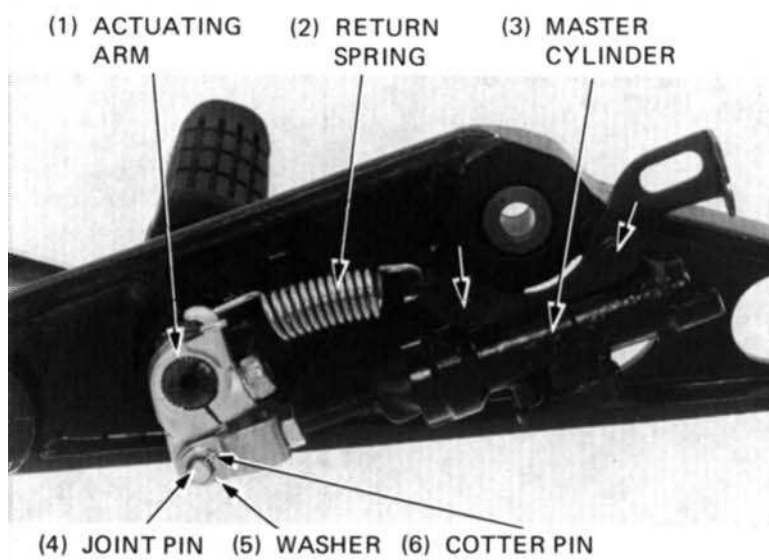
Remove the rear brake switch from the right footpeg bracket and unhook the switch spring from the rear brake return spring.



Remove the hose connector screw and disconnect the reservoir joint hose from the master cylinder.



Unhook the rear brake return spring from the actuating arm.
Remove the actuating arm from the pedal shaft.
Remove the cotter pin, washer and joint pin, and disconnect the actuating arm from the master cylinder push rod.
Remove the master cylinder from the footpeg bracket.

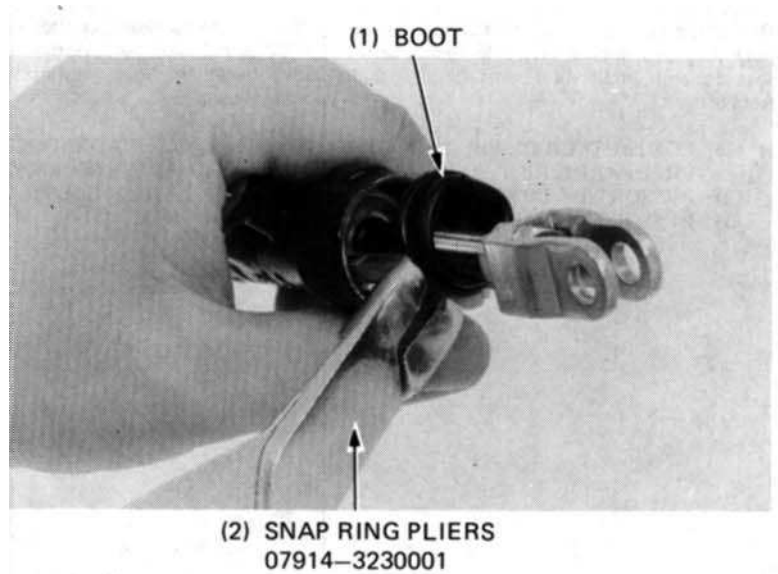




15. Hydraulic Brakes

DISASSEMBLY

Remove the rubber boot. Remove the snap ring and push rod from the master cylinder body. Remove the master piston, primary cup and spring. It may be necessary to apply a small amount of air pressure to the fluid outlet to remove the master piston and primary cup. Clean the piston and inside of the master cylinder with brake fluid.

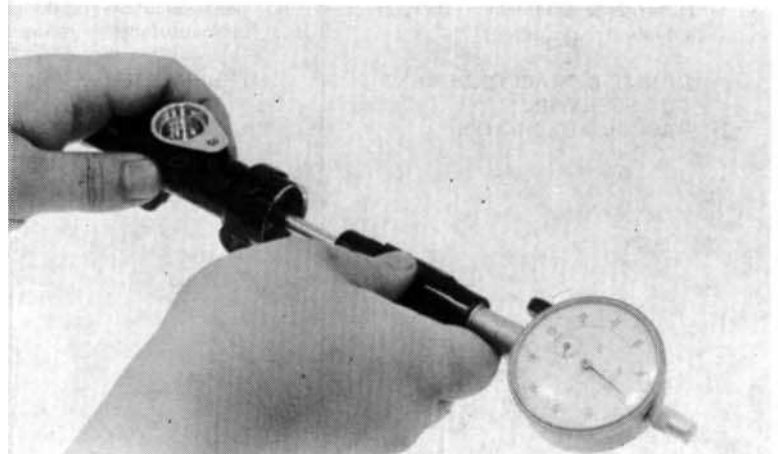


CYLINDER I.D. INSPECTION

Measure the inside diameter of the master cylinder bore.

SERVICE LIMIT: 14.06 mm (0.554 in)

Check for scores, scratches or nicks.

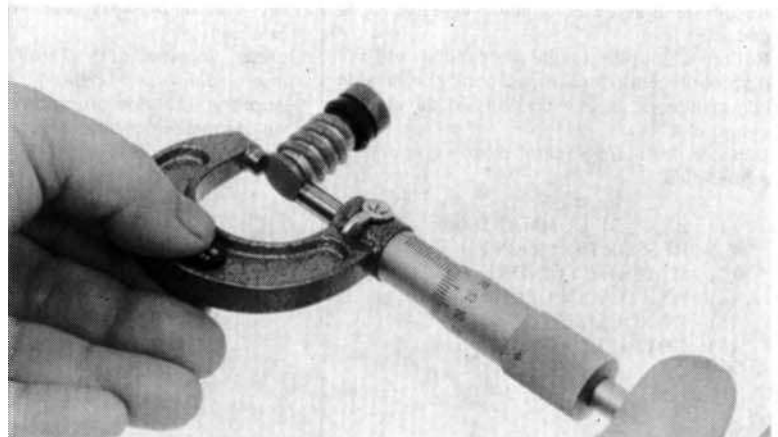


PISTON O.D. INSPECTION

Measure the master piston O.D.

SERVICE LIMIT: 13.95 mm (0.549 in)

Check the primary cup and piston cup for damage.





ASSEMBLY

CAUTION

Keep the master cylinder piston, cylinder and spring as a set; do not substitute individual parts.

Assemble the master cylinder. Coat the master piston, primary and secondary cups with clean brake fluid before assembly. Install the spring and primary cup together. Install the piston. Install the push rod and snap ring.

CAUTION

Do not allow the lips of the cups to turn inside out and be certain the snap ring is firmly seated in the groove.

Install the rubber boot.

INSTALLATION

Install the master cylinder onto the footpeg bracket.

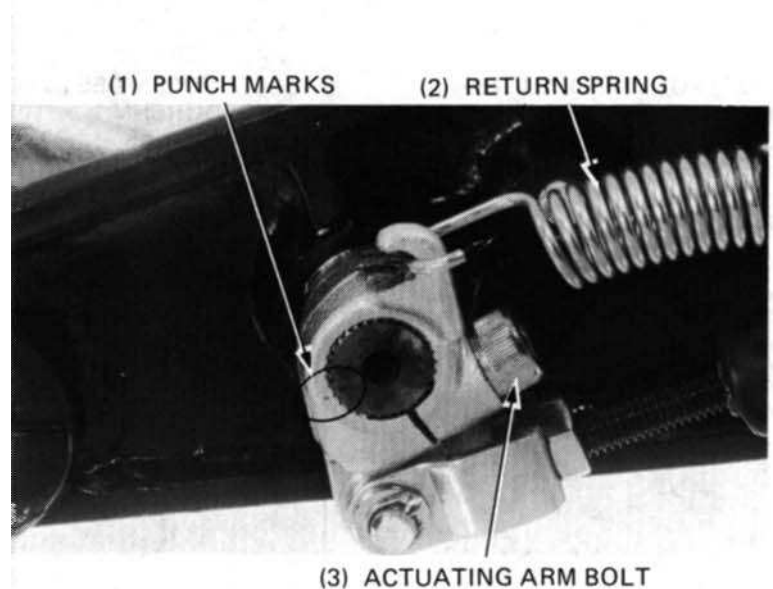
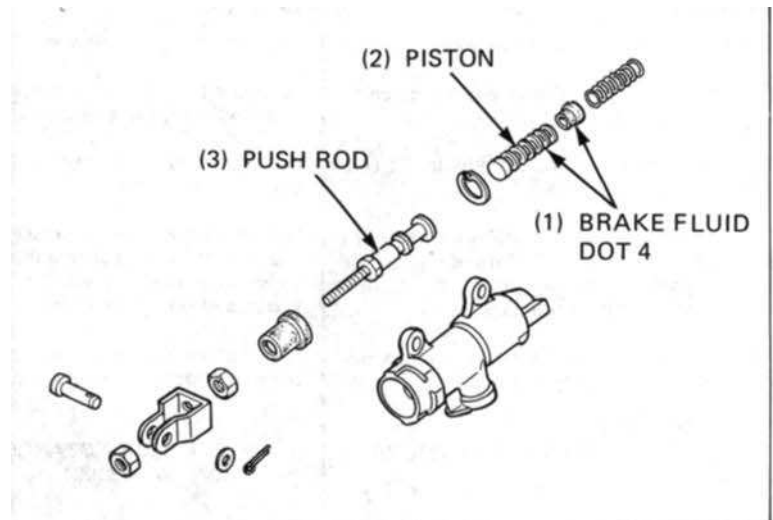
Connect the actuating arm to the master cylinder push rod with the joint pin, washer and a new cotter pin.

Install the actuating arm onto the pedal shaft, aligning the punch marks on the arm and shaft.

Tighten the actuating arm bolt.

Hook the rear brake return spring to the actuating arm.

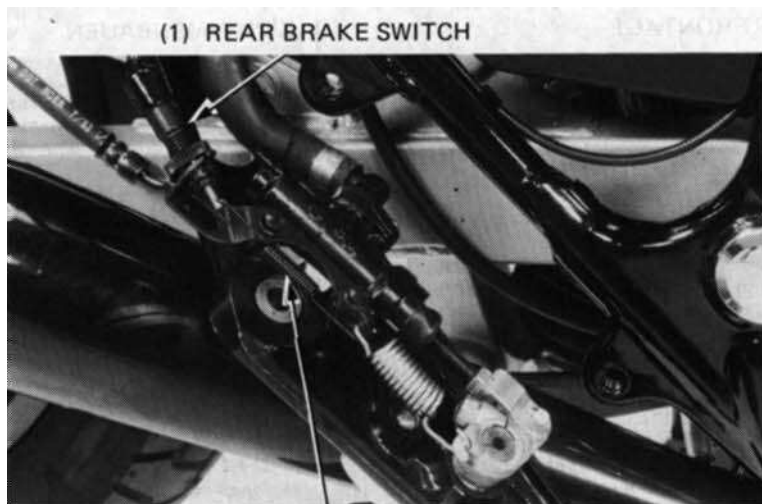
Connect the reservoir joint hose to the master cylinder with a new O-ring and the screw.





15. Hydraulic Brakes

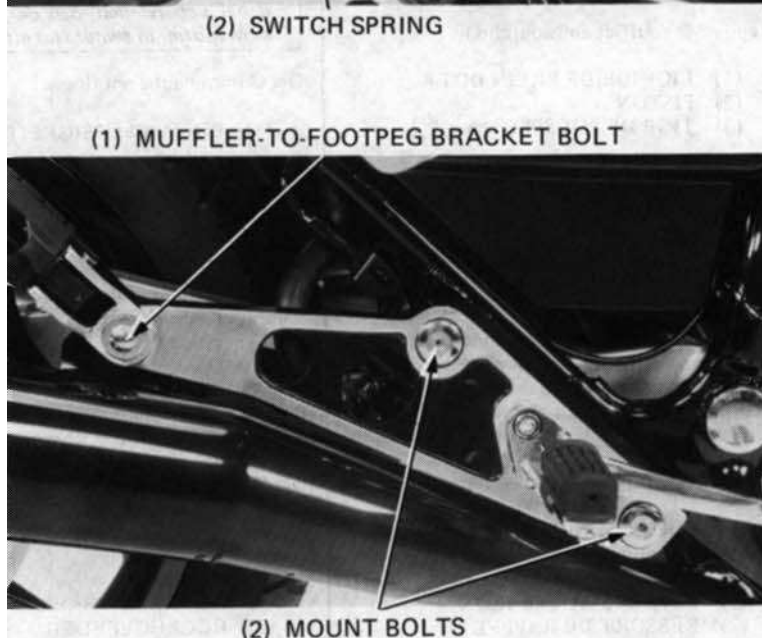
Hook the rear brake switch spring to the return spring and install the switch to the footpeg bracket.



Install the footpeg bracket to the frame and tighten the mount bolts.

TORQUE: 35-45 N.m
(3.5-4.5 kg.m, 25-33 ft.lb)

Install the muffler-to-footpeg bracket bolt and nut.



Connect the brake hose to the master cylinder with the oil bolt and two sealing washers, and tighten the oil bolt.

TORQUE: 25-30 N.m
(2.5-3.0 kg.m, 18-22 ft.lb)

Fill and bleed the rear brake system (page 15-4).

Adjust the rear brake switch (page 3-12).

